ANWAR KADER

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EDUCATION

The University of Chicago — Chicago, Illinois

Bachelor of Sciences in Computer Science, Bachelor of Arts in Economics | GPA: 3.88/4.0 | Dean's List — June 2028

- Coursework: Foundations of Distributed Systems, Theory of Algorithms, Systems Programming, Mathematical Foundations of Machine Learning, Introduction to Cryptography

EXPERIENCE

Software Engineer Intern — June 2026 – August 2026

Infleqtion — Chicago, Illinois

- Led development of a React/TypeScript web platform for a quantum-computing cloud service used by 100+ customers, shipping 13 routes backed by a typed API client spanning **34 methods over 17 REST endpoints**
- Set client types from backend OpenAPI spec, turning FastAPI schema drift into compile-time, instead of runtime, errors
- Built a client-side caching layer (promise memoization + 5-entry LRU) that removes an entire API round trip from route navigation, cutting return-navigation render by **80% (310ms to 60ms)** via zero network calls on cached views
- Moved circuit rendering to a Web Worker, keeping UI responsive on large circuits that previously blocked the main thread
- Hardened the browser security layer: added an 11-directive CSP plus HSTS and clickjacking headers, and sanitized every HTML/SVG injection point with DOMPurify
- Authored 203 Vitest unit tests and 9 Playwright E2E specs covering auth, caching, and API error paths, gated in GitHub Actions CI across 3 Node versions

Visualization Software Engineer Intern — September 2025 – May 2026

Research Computing Center's Data Visualization Laboratory — Chicago, Illinois

- Optimized system performance by reducing application load time by over **95% (20 minutes to 40 seconds)** through **object pooling**, efficient memory management, and minimizing costly instantiations of 3D objects.
- Extracted data from large NetCDF4 global climate datasets **(40+ GB)** using **Python** (XArray, Pandas), converting **10+ atmospheric variables** into optimized formats for 3D visualization.
- Presented application to **NASA scientists** and over 15 computational scientists**,** including the center's director

Research Assistant Intern — April 2025 – December 2025

SUPERgroup Research Collective — Chicago, Illinois

- Greenfield development of a Python pipeline that prompts the Claude API against a 29-category PII taxonomy to detect and redact identifiers across **7,500+** speech-transcript utterances, supporting an ongoing PhD research project and paper
- Cut redaction-review page latency **77%** by replacing per-term full-column regex scans over a pandas DataFrame with hash-indexed occurrence maps built in a single load-time pass, plus version-stamped memoization and table pagination
- Developed an automated pipeline in Python to assist in labeling and analyzing **over 1,000 data entries** from a dataset of 3,000+, collapsing multiple 2hr+ manual rounds of review into single 2-minute runs while maintaining data integrity

PROJECTS

Media Viewing Platform | React, AWS [mediaviewer.vercel.app]

- Utilized the Google YouTube API to import user subscriptions and created a personalized homepage that retrieves latest content from **over 100 channels daily** to each user. Implemented caching, channel tags, and a watch later system.
- Engineered an **Agent Harness** for AWS Bedrock with LangChain, providing custom tools to access MongoDB and Lambda functions to create video playlists on demand in **under 15 seconds**
- Chose Lambda functions and MongoDB database so page loads never hit the upstream YouTube API, **reducing latency by 70%,** and minimized total upstream API calls due to being constrained by YouTube API quota

Justice: Automated Legal Form Creation: Top 5 at ALI Builds Hackathon | devpost.com/software/justice-in-a-flash]

- Built a **3-agent AI review pipeline** (drafter, opposing counsel, judge) generating 3 independent perspectives per case, by chaining Claude Sonnet 4.6 calls with **shared case-fact context** across the API layer.
- Enabled 100% offline functionality with 0 required API keys, by writing **typed mock-fallback paths** for every API external call (vision, STT, classification, drafting) behind a shared client layer.

MCP Server for Flight API | Python, MCP [huggingface.co/spaces/WoothAmwar/mcp-sentiment]

- Built and hosted a **custom MCP Server** integrating the Amadeus Flights API to retrieve flight details between locations.

- Used Python and Gradio, deployed on HuggingFace and integrated with custom MCP Client and Claude Desktop to enable real-time flight data queries within user conversations

Schoolwide Game Tracking Platform | React, MongoDB [github.com/GoldEnter21/NCSSM-Spoons-Online]

- Shipped website to organize our school's student-led game of tag with **over 250 registered participants**, giving them the ability to track player eliminations, check user information, and view a live leaderboard.
- Created with the React library for the frontend, MongoDB to store information, and Vercel to host the application.

LEADERSHIP

University of Chicago ACM Board Member February 2025 – Present

- Lead the Software Engineering Committee, organizing weekly meetings for the over 70 members to teach data structures, algorithms, and technical interview preparation, enhancing members' career readiness and coding skills.

TECHNICAL SKILLS & CERTIFICATIONS

- **Languages and Tools:** Python, JavaScript, TypeScript, C | Claude Code, Codex
- **Frameworks and Databases:** React, Node.js**,** MongoDB, PostgreSQL
- **Backend & Cloud**: AWS (Lambda, API Gateway, EC2, VPC), GCP, Vitest, Playwright | AWS Certified AI Practicioner